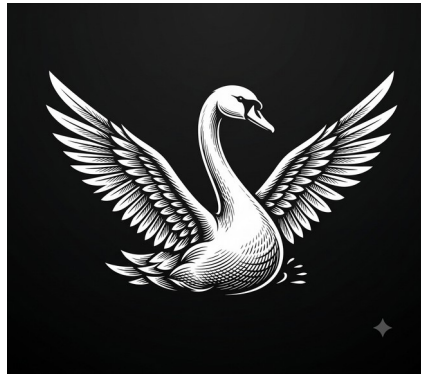




Hamsa

Neera Kshira Nyaya — Separating Milk from Water, One Scan at a Time.

PRODUCT REQUIREMENTS DOCUMENT

A consumer-facing platform for detecting counterfeit and substandard medicine in India

Hackathon track: Counterfeit & Substandard Medicine Detection

Written to be handed directly to an AI coding assistant and built from.

Contents

1. Executive Summary
2. The Problem
3. Goals, Non-Goals & Success Metrics
4. Target Users
5. Competitive Landscape
6. Differentiators
7. Brand & Naming
8. Product Flow — How It Works
9. The Evidence Engine
10. Verdict Levels
11. Security, Privacy & Compliance
12. Accessibility & Reach
13. Batch Timeline & Regulator Investigation View
14. System Architecture
15. Data Model
16. API Specification
17. Screens & UX Specification
18. Technology Stack
19. Build Plan (34 Hours)
20. Testing & QA Checklist
21. Deployment Guide
22. Demo Script for Judges
23. Business Model & Sustainability
24. Honest Limitations — What We Are Not Claiming
25. Elevator Pitch
26. Sources Referenced
27. Pre-Submission Checklist

1. Executive Summary

Hamsa is a consumer-facing platform that helps people in India tell a genuine medicine pack from a counterfeit or substandard one — not by trusting a single QR code or a single AI-generated score, but by independently cross-checking several sources of evidence and showing exactly which one triggered each warning.

Every month, CDSCO (the Central Drugs Standard Control Organisation) publishes lists of medicine batches that failed quality testing — Not-of-Standard-Quality (NSQ) — or were found to be outright spurious. This is real, government-verified evidence of unsafe medicine already circulating in the market. It is published as static PDFs that almost no patient, pharmacist, or shopkeeper reads before a specific batch reaches their hands.

Hamsa turns that evidence into an instant, sourced, actionable answer. A user scans a medicine's QR/DataMatrix code, uploads a photo of the package, or types the batch number by hand — manual entry is treated as the reliable default, not a fallback, because OCR on Indian pharmaceutical packaging is genuinely hard. Hamsa's Evidence Engine then independently checks: whether the batch appears in CDSCO's NSQ/spurious alerts, whether the printed label agrees with what is encoded in the code, whether the medicine has expired, and — as a clearly labeled preview — whether the code shows signs of being cloned or replayed. Every check is shown as its own row in an Evidence Matrix, every row is traceable to its source through Evidence Provenance, a Safe Action layer turns the verdict into plain guidance, and a stated Conflict Resolution policy governs what happens when checks disagree. The verdict is never a black-box number, and a manufacturer or pharmacy wrongly flagged has a real, documented way to dispute it.

Hamsa reaches people over WhatsApp in their own language, with no app install and no literacy requirement, and is designed to keep working — on cached data — when the connection doesn't.

What this document is for

This is a single, self-contained build specification. It carries the product rationale, the full data model, the API contract, and a screen-by-screen UX description, so it can be handed to an AI coding assistant (or a human engineer) and built from directly, without needing to consult any other document.

2. The Problem

2.1 The structural weakness in "scan a code, get a score"

The common framing for this problem — photograph a medicine package, read the batch and label via OCR, check the QR/DataMatrix code against a manufacturer registry, look for print anomalies, and return a single 0–100 trust score — has a weakness worth fixing before anything else is built:

- A QR/DataMatrix code proves the code was generated correctly, not that the specific pack in someone's hand is genuine. A real, valid code can simply be photographed and reprinted onto a fake box — this is the single biggest hole in "scan and verify" systems, and Hamsa names it explicitly rather than implying it away (see Section 9.1 and Section 24).
- A single trust score is a black box. In a domain where a wrong answer can cause real harm, both judges and real users will ask "based on what, exactly?" — and a black box has no answer.
- India already mandates traceable codes, but verification is fragmented. GS1 DataMatrix codes carrying batch, expiry, and manufacturer data are mandatory on India's top drug brands under Rule 96 of the Drugs Rules, 1945 (inserted by G.S.R. 823(E), notified 17 November 2022, in force from 1 August 2023) — but each code today resolves only to its own manufacturer's system, if it resolves anywhere at all.
- CDSCO already collects hard evidence of bad batches every month, through its NSQ and spurious-drug alerts — published as PDFs, not as a searchable consumer warning system.
- Government compliance data is itself opaque: in a written reply to the Lok Sabha on 1 August 2025, the government confirmed it has not disclosed how many of the mandated manufacturers have actually adopted the QR code system, more than two years after the rule took effect. That opacity is itself an argument for an independent verification layer.

2.2 Scale — current, India-specific evidence

The most-cited figure for this problem is WHO's 2017 global surveillance estimate — roughly 1 in 10 medical products in low- and middle-income countries (10.5% of 48,218 sampled products across 88 countries) is substandard or falsified, at an estimated cost of US\$30.5 billion a year. That figure is nine years old by the time of this pitch. Hamsa's own dataset provides a sharper, current, India-specific number: CDSCO's monthly alert for June 2025 alone flagged 189 drug batches as Not-of-Standard-Quality or spurious — 130 by state labs, 55 by central labs, and 4 confirmed spurious — across 15 state and central testing laboratories, with Karnataka's state labs alone accounting for 53 of them. This is not a hypothetical baseline; it is the live dataset Hamsa ingests every month.

2.3 Regulatory context Hamsa must build inside

- Rule 96, Drugs Rules 1945 (Schedule H2) — inserted by G.S.R. 823(E), 17 November 2022; mandatory QR/barcode on the top 300 drug brands by sales value, in force from 1 August 2023.
- G.S.R. 506(E), the Drugs (Seventh Amendment) Rules, 2026 — notified 22 June 2026, expanding Schedule H2 to all vaccines, all antimicrobials, all NDPS narcotic/psychotropic drugs, and all anti-cancer drugs. Rollout is staggered, not simultaneous: vaccines, anti-cancer, and NDPS drugs from 1 July 2027; antimicrobials from 1 July 2028.
- Digital Personal Data Protection Act, 2023 (DPDP Act) and the Digital Personal Data Protection Rules, 2025 — the Rules were formally notified by MeitY on 13 November 2025, operationalising the Act (in force since August 2023). Compliance is phased over 18 months from notification, but core principles — consent, purpose limitation, data minimisation, security safeguards — are the framework Hamsa is designed inside from day one, not retrofitted later (see Section 11.3).

3. Goals, Non-Goals & Success Metrics

3.1 Goals

- Turn CDSCO's NSQ/spurious data into a searchable, instant, sourced consumer answer.
- Make every verdict auditable — a person, a judge, or a regulator can always see exactly which check produced it and where the underlying fact came from.
- Reach rural, low-literacy, low-connectivity users with no app install and no English requirement, primarily via WhatsApp.
- Give a wrongly-flagged manufacturer or pharmacy a real, documented way to dispute a verdict.

3.2 Non-goals — stated up front, to keep the build honest

- Hamsa is not a lab-grade forensic authentication tool. It does not claim to detect physical counterfeiting artifacts from a phone photo.
- Hamsa is not a replacement for a pharmacist's or doctor's advice.
- Hamsa is not a production track-and-trace system with manufacturer-side serialisation deactivation at the point of sale. That requires direct manufacturer integration, which is out of reach for a hackathon build and is named as a roadmap item, not a current capability (see Section 24).

3.3 Success metrics for the hackathon build

- $\geq 90\%$ successful QR/DataMatrix decode rate on the prepared demo pack set.
- $\geq 85\%$ OCR field-extraction accuracy (manufacturer, batch, expiry) on a 20-pack sample, with manual correction available for the rest.
- End-to-end scan-to-verdict latency under 5 seconds on the web dashboard, under 15 seconds over WhatsApp.
- Zero unhandled exceptions during the rehearsed live demo run-through.

- 100% of Evidence Matrix rows show a non-empty source and confidence value — no row is ever shown with nothing behind it.

4. Target Users

Primary — the consumer

A price-sensitive person buying over-the-counter or prescribed medicine at a pharmacy in a tier-2/3 town or village. Moderate-to-low digital literacy, owns a basic smartphone with WhatsApp installed, connectivity is unreliable. This person will never download a dedicated app for a one-off check — WhatsApp is the product's real front door.

Secondary — the pharmacist / chemist

Wants a fast second check before dispensing a batch, especially one from a less familiar distributor. Uses the web dashboard, values speed over depth.

Tertiary — the drug inspector / regulator (prototype only)

Wants to see open reports and cross-manufacturer patterns in one place. This role exists only as a clearly labeled, unauthenticated-in-name-only prototype screen behind a basic access gate (see Section 11.1 and 13.2) — it is not a production regulator tool.

5. Competitive Landscape

The authentication space is more crowded than a first pass usually finds. This table is deliberately unflattering to Hamsa where it should be.

Player	What it does	Where it falls short
PharmaSecure (psVerify)	Manufacturer-enrolled serialised codes; SMS/app verification. Deployed on over 1.2 billion packages.	Only covers enrolled manufacturers. Reported active mobile usage is very small for a decade-old, well-funded incumbent — real evidence that a bare verification app struggles for engagement even at scale.
Zydus Verify (Zydus Cadila)	A large Indian pharma company's own scratch-code verification app.	Single-manufacturer only. Confirms big pharma is building this in-house too.
CDSCO's own QR/DataMatrix mandate	Legally mandates serialised codes on an expanding set of drug categories (Section 2.3).	No unified government verification app or public-warning layer. Compliance data is not even publicly disclosed (Section 2.1).
Scantrust / Acviss / Qrynto / Uniqode	Market "impossible-travel" clone/replay detection.	Real technology, but not a new idea — Hamsa's version is explicitly labeled a preview, not a claim of parity.
Karnataka FSDA real-time NSQ portal	Launched May 2026: automatically freezes NSQ-flagged batches at stockists and retailers, tracks NDPS medicine sales in real time.	Trade-side, not consumer-facing — a pharmacy's stock gets frozen, but the person already holding the pack at home has no way to check it. The most concrete government threat found, and it is live today, not hypothetical.

Player	What it does	Where it falls short
Kerala "MedWatch" (state government)	Reported to be in development by Kerala's State Drugs Control Department — a citizen-facing app integrating pharmacy APIs.	State-specific, and as of this writing there is no confirmed public launch — the most direct long-term threat, but not yet a competitor in the market.
MediCheck / HealthSewa (hackathon prototypes)	Scan medicine, read alerts, single-API verification, rural framing.	No evidence of a real government NSQ data pipeline or a stated policy for conflicting evidence behind either one.
Blockchain track-and-trace pilots	Multiple India and global pilots.	Adoption stalls on governance and legacy-system integration, not the ledger technology.

Table 5.1 — Competitive landscape, including two findings a shallower pass would have missed: the live Karnataka FSDA portal, and psVerify's real-world adoption numbers.

What none of the players above are confirmed to have, together, is: a real structured CDSCO NSQ/spurious pipeline; a transparent Evidence Matrix showing every check and its source; and an explicit, stated policy for what happens when signals disagree, backed by a real dispute process. That combination — not "we built a medicine scanner" — is the actual claim.

6. Differentiators

1. Real-time NSQ/spurious ingestion, structured for consumers

CDSCO's monthly NSQ and spurious-drug lists (central and state labs) are parsed into a searchable database: batch number, product, manufacturer, failure reason, alert month, and a link back to the source document.

2. The Evidence Engine — explainable by construction, not by label

Every verdict is backed by an Evidence Matrix (9.2), each row traceable through Evidence Provenance (9.3), translated into plain guidance through the Safe Action layer (9.4), and reconciled through a stated Conflict Resolution policy (9.5) when signals disagree. This is the direct answer to "how is this different from a trust score" — a mechanism, not an assurance.

3. The clone-code weakness is named, not hidden

A static QR/DataMatrix code can be photographed and reprinted onto a fake box, and the label-vs-code check alone will not catch it, because both sides will agree. Hamsa says this out loud (Section 9.1, Section 24) instead of implying the problem is solved, and offers the clone/replay preview as a clearly labeled, honestly-scoped answer to it.

4. A label-vs-code consistency check grounded in the actual rule

CDSCO's mandate requires printed text to match the encoded data. Comparing OCR'd label text against the decoded QR/DataMatrix payload — using normalised, fuzzy matching against a manufacturer-alias table, not brittle exact string equality — is a legitimate, rule-grounded fake signal.

5. A real dispute process

A manufacturer or pharmacy wrongly caught by an escalated conflict, or by a stale NSQ record, has a documented way to contest it (Section 9.6) — not just a promise that "corrections are shown, never hidden."

6. Built to be legal, not just clever

Consent, data minimisation, and retention limits are designed in from the start against the DPDP Act/Rules (Section 11.3), not treated as a post-hackathon problem.

7. Reach

A WhatsApp bot, manual entry as the default (not a fallback), and regional-language replies — including voice notes — mean verification never requires an app install or literacy in English or in reading at all.

7. Brand & Naming

The name

Hamsa is the Sanskrit swan (or goose) of Indian philosophy, traditionally credited with the ability to drink only the milk from a mixture of milk and water, leaving the water behind — neera-kshira-nyaya, "the logic of milk and water," a classical metaphor for viveka, discernment between the real and the false. It is the vahana (vehicle) of Saraswati, the goddess of knowledge. The job here is not protection, it's discernment — telling the genuine pack from the counterfeit one when both look identical on a shelf.

Tagline

"Neera Kshira Nyaya — Separating Milk from Water, One Scan at a Time."

Brand disambiguation — say this before the pitch, every time

"Hamsa" is also the name of an unrelated protective hand-shaped amulet (also spelled "khamsa") from Middle Eastern and North African tradition — a different word, symbol, and cultural origin that happens to share a transliteration. Always introduce the name together with the swan and the milk-water story, and keep the logo unambiguously a bird, never a hand shape. The closest sound-alike in the broader "X-AI" naming space is Hamming AI, an unrelated funded voice-AI startup — low real risk, but a reason not to force an "-AI" suffix onto "Hamsa" itself. This is a quick naming check, not trademark clearance — worth one more look before any public submission beyond the hackathon.

Logo

An abstract swan in flight-ready posture, checkmark motif across the body for "verified." Delivered as an editable vector plus PNG exports at multiple sizes; the icon on this document's cover is the working mark.

8. Product Flow — How It Works

1. **Capture.** Manual batch-number entry is the reliable default path, always available and never hidden behind a scan attempt. Scanning the QR/DataMatrix code or uploading a package photo is the fast path when the pack, lighting, and print quality cooperate. The user can always fall back to manual entry mid-flow, and an obvious OCR misread can always be corrected by hand before submission.
2. **Extract.** Decode the QR/DataMatrix payload and OCR the printed label. Extracted manufacturer and product names are normalised (case, punctuation, common abbreviations) and resolved against a manufacturer-alias table before any comparison is made — see Section 9.1.
3. **Cross-check.** Compare printed fields against decoded fields; look the batch up against the NSQ/spurious database; check the expiry date against today's date.
4. **Preview pattern check.** Check the batch/serial against a synthetic scan-log dataset for impossible-travel patterns — clearly labeled as a preview, on synthetic data, every time it is shown.
5. **Evidence Engine.** Combine every signal through the Conflict Resolution policy into one verdict, and build the Evidence Matrix behind it.
6. **Explain and guide.** Show the Evidence Matrix, let the user drill into Provenance, and surface the Safe Action text in their language — as text, and as a voice note on WhatsApp.

Offline and low-connectivity behaviour

If the device has no network at the moment of a scan, Hamsa does not simply fail. The web and WhatsApp clients keep a locally cached snapshot of the most recent NSQ/spurious dataset (refreshed whenever connectivity is available) and can return a verdict against that cached snapshot immediately, with a visible banner stating the snapshot's "last updated" timestamp. The scan is queued and silently re-verified against live data the next time the device reconnects; if the re-check changes the verdict, the user is notified. This exists specifically because the target user — rural, low-connectivity India — is exactly the person a "requires live internet" design would fail first.

9. The Evidence Engine

9.1 Decision logic

Pseudocode for the core decision path. Every branch produces a labeled, sourced reason — never a single number standing alone. Manufacturer/product name comparisons use normalise() + fuzzy match, not exact equality, so that formatting differences ("Cipla Ltd." vs "CIPLA LIMITED" vs a Devanagari rendering of the same name) are not mistaken for evidence of tampering.

```
evidence = []

# Official record – always checked first
if batch in nsq_or_spurious_database:
    evidence.append(Signal("nsq_status", "FAIL", RED, source_document, alert_date))

# Expiry is its own, independent signal – never folded into the counterfeit verdict
if expiry_date and expiry_date < today():
    evidence.append(Signal("expiry_status", "EXPIRED", GREY, "Decoded / OCR expiry field"))

# Label-vs-code, using normalised + fuzzy matching, not exact string equality
if not fuzzy_match(normalise(decoded_fields.manufacturer),
normalise(ocr_fields.manufacturer),
alias_table=manufacturer_aliases, threshold=0.85):
    evidence.append(Signal("label_consistency", "FAIL", AMBER, "OCR vs decoded payload"))

if qr_payload is malformed or missing required fields:
    evidence.append(Signal("code_validity", "FAIL", AMBER, "Decoded payload"))
```

```

if preview_mode and scan_pattern_is_impossible(batch, history):
    evidence.append(Signal("replay_pattern", "INCONCLUSIVE", AMBER, "[Preview] synthetic
data"))

dispute = lookup_open_dispute(batch_number)    # Section 9.6

verdict = resolve_conflicts(evidence)          # Section 9.5
matrix = build_evidence_matrix(evidence)
action = safe_action_for(verdict, evidence)    # Section 9.4

show(verdict, matrix, action, dispute_banner=dispute, language=user.language)

```

9.2 Evidence Matrix — the central UI element

Every independent check is shown as its own row, never collapsed into one score. Worked example — a real NSQ match:

Signal	Status	Evidence	Source
Code validity	Pass	DataMatrix decodes correctly; all required fields present	Decoded from scanned code
Label-code consistency	Pass	Printed manufacturer matches encoded manufacturer (fuzzy match, 0.97 confidence)	OCR vs decoded payload
Expiry status	Pass	Expiry 03/2028 — not yet expired	Decoded payload
Official NSQ/spurious status	FAIL — RED	Batch AX2291 listed Not of Standard Quality (dissolution failure)	CDSCO NSQ list, March 2026, Central Labs
Replay/clone pattern [Preview]	Inconclusive	No live scan-log infrastructure in this build	Synthetic demo data only
Visual heuristic	Pass	No blur, crop, or missing print fields detected	Uploaded package photo
Community reports	None (regulator-view only)	No confirmed reports filed for this batch	User-submitted, unverified

Table 9.1 — Overall verdict: RED — Stop (driven by the NSQ row; Conflict Resolution rule 1).

A second worked example — an expired, otherwise clean pack:

Signal	Status	Evidence	Source
Code validity	Pass	Decodes correctly	Decoded from scanned code
Label-code consistency	Pass	Fields match	OCR vs decoded payload
Expiry status	FAIL — Expired	Expiry 07/2025 — expired 14 months ago	Decoded payload
Official NSQ/spurious status	Pass	No match found in current CDSCO alerts	CDSCO NSQ/spurious database

Table 9.2 — Overall verdict: Expired (Section 10) — a distinct outcome from counterfeit RED, with its own Safe Action text.

9.3 Evidence Provenance

Every Evidence Matrix row expands into:

- **Source type** — government record / decoded code / OCR reading / user photo / community report.
- **Reference** — the source document name or scan reference.

- **Confidence** — High / Medium / Low, mapped from extraction_confidence for OCR'd PDF data, or "Structured" for decoded fields.
- **Last updated** — when that record was ingested or refreshed.
- **Dispute status** — none / open / resolved, surfaced directly on the relevant row so a viewer never has to guess whether a fact is currently contested (Section 9.6).

This is what separates an evidence report from an AI opinion — every fact traces to where it came from, when it was last confirmed, and whether anyone has formally challenged it.

9.4 Safe Action layer

Verdict / reason	Action shown to the user
RED — NSQ/spurious match	Do not consume. Show this result to a pharmacist. Keep the packaging. [Report this batch]
RED — label-code mismatch (high confidence) / escalated conflict	Do not consume. This pack shows signs of tampering or inconsistency. Keep the packaging and report it.
Expired (independent of counterfeit status)	Do not use — this pack expired on [date]. Dispose of it safely. No report needed unless you also suspect tampering.
AMBER — single signal	Don't discard yet — confirm with your pharmacist before use. [Report] [Re-scan]
AMBER — escalated (two independent signals)	Treat with high caution. Confirm with a pharmacist and consider reporting.
GREEN — no warning found	No warnings found in our records. This does not confirm authenticity — always buy from a licensed pharmacy.
Any verdict with an open dispute on this batch	This result is under review — a manufacturer or pharmacy has formally disputed part of this record. The verdict above still applies until the review concludes.

9.5 Conflict Resolution

1. **Official evidence always wins.** A confirmed CDSCO NSQ/spurious match sets the verdict to RED regardless of what any other signal shows.
2. **One non-official signal alone caps at AMBER.** A single label mismatch, visual heuristic, or preview pattern is never enough to declare RED by itself — one weak signal is not proof.
3. **Two independent non-official signals escalate to RED.** The two signals must come from genuinely different evidence categories (e.g. label-code mismatch AND a preview replay pattern), not two symptoms of the same underlying OCR error. Escalations are logged and shown distinctly in the Evidence Matrix, so a RED from escalation never looks identical to a RED from an official record.
4. **Corrections are shown, never hidden.** If an NSQ listing is later corrected or superseded, the verdict always reflects the most recent record, but the full history stays visible in Provenance and the Batch Timeline.
5. **Unverified community reports never produce RED or AMBER alone, and are never shown publicly.** They surface only inside the authenticated Regulator Investigation View (Section 13.2), pending confirmation — this is what stops the product from enabling mob-flagging or unverified public accusations against a pharmacy or manufacturer, and from being used to reverse-engineer a competitor's distribution footprint.
6. **Expiry is evaluated independently of counterfeit status.** A pack can be both expired and NSQ-flagged; when that happens, RED (the official record) is shown as the primary verdict with "also expired" as a secondary tag, and the Safe Action text for RED is shown, not the milder expired-only text.
7. **An open dispute never silently changes the verdict.** It adds a visible "under review" banner (Section 9.4) and pauses new escalations built on the specific disputed signal until the dispute resolves (Section 9.6) — the user still sees the current evidence-based verdict, not a blank screen.

9.6 Manufacturer / Pharmacy Dispute & Correction Process

Any manufacturer or pharmacy that believes a batch has been wrongly flagged can submit a dispute through a dedicated form (POST /api/disputes, Section 16), attaching supporting evidence (a certificate of analysis, a corrected NSQ notice, proof of a data-entry error). For the hackathon build, submissions are verified against a simple registered-manufacturer email domain check — explicitly labeled prototype-grade, with a real verification workflow (e.g. GST/drug-licence number lookup) named as the production requirement. Every open dispute is shown as a visible banner on the affected batch's verdict screen (never hidden), reviewed against a stated target SLA of five business days for a production deployment, and resolved as either upheld (verdict stands) or overturned (verdict and Evidence Matrix are corrected, with the correction itself kept visible in the Batch Timeline per Conflict Resolution rule 4). Unresolved disputes older than the SLA window are surfaced to the Regulator Investigation View for manual escalation.

10. Verdict Levels

- **RED — Stop:** a sourced NSQ/spurious match, a clear label-vs-code mismatch, or an escalated conflict (Section 9.5).
- **AMBER — Check further:** a single non-official signal, incomplete data, or a preview-flagged pattern.
- **Expired:** its own outcome, independent of counterfeit status, with its own Safe Action text (Section 9.4) — never presented identically to a counterfeit RED.
- **GREEN — No warning found:** explicitly not "genuine," only "nothing in our data says otherwise."

11. Security, Privacy & Compliance

This section exists because the previous draft of this product had none of it. A healthcare-adjacent consumer app collecting photographs, phone numbers, and location data in India in 2026 has to answer these questions before a judge asks them.

11.1 Authentication & authorization

Endpoint group	Access model
Read-only public (/verify, /lookup-batch, /batch-timeline, /disputes GET)	Open, no login required — this is the product's entire value proposition — but rate-limited per Section 11.2.
/report (POST)	Requires a lightweight identity check: the reporting WhatsApp number, or a CAPTCHA/Turnstile challenge on web. No account creation required.
/disputes (POST)	Requires a registered-manufacturer email domain match (hackathon-grade). Production requirement: GST/drug-licence number verification.
/regulator/* (all)	Requires a signed, short-lived regulator token (a simple shared-secret + username login for the hackathon prototype). Explicitly labeled prototype-grade — production requires real role-based access control, named as a roadmap item, never silently assumed to be present.
/user/data (DELETE)	Requires the same WhatsApp-number identity check as /report — this is the person's own data, so proof of the same number is sufficient.

11.2 Abuse prevention & rate limiting

Endpoint	Limit
POST /api/verify	30 requests / minute / IP; 200 / day / IP
POST /api/report	5 / day / reporting identity
POST /api/disputes	10 / day / manufacturer domain
WhatsApp bot inbound messages	1 verdict lookup / 10 seconds / phone number, to absorb accidental double-sends without feeling throttled

All uploaded package photos are validated for file type and size (JPEG/PNG/WebP, 10 MB max) before processing, and EXIF metadata — including embedded GPS coordinates — is stripped on ingestion, before the file is ever stored. This closes a location-leak that a raw-upload pipeline would otherwise create silently.

11.3 Data protection — designed against the DPDP Act, 2023 and DPDP Rules, 2025

- Consent notice shown on first interaction (web banner; first WhatsApp message), stating in plain language what is collected, why, and for how long, before any photo or phone number is processed.
- Data minimisation: package photos are processed for OCR/evidence purposes and retained for 30 days by default, then automatically deleted — unless the user opts in to keep scan history, or the photo is attached as evidence to an open report or dispute, in which case it is retained until that case resolves.
- Location data (scan_events.approx_location) is stored at pincode/district granularity by default, never as precise GPS coordinates, and only when the user has opted into the clone/replay preview feature.
- Data-principal rights are honoured through a plain-language command: replying DELETE MY DATA on WhatsApp, or a "Delete my data" button on web, triggers DELETE /api/user/data (Section 16), which erases the user's scan history and any photos not attached to an open case.

- No data leaves India-hosted infrastructure without being explicitly documented — the hackathon build's hosting choices (Section 18) are noted with this in mind.

11.4 Image & data retention policy

Data	Default retention
Uploaded package photos (not attached to a report/dispute)	30 days, then auto-deleted
Uploaded package photos (attached to an open report/dispute)	Until the case resolves, plus 90 days
Scan history (batch numbers, verdicts)	Kept only if the user opts in; otherwise not persisted beyond the session
scan_events (preview/synthetic location data)	Pincode/district granularity only; 180 days
Consent log	As long as the account/identity is active, per DPDP record-keeping expectations

11.5 Legal disclaimers & liability posture

Shown persistently in the UI, not buried in a terms page: Hamsa is not a substitute for professional medical or pharmacist advice. A GREEN verdict is not a certificate of authenticity — it means nothing adverse was found in the data checked. A RED verdict is based on the evidence shown in the Evidence Matrix, and any manufacturer or pharmacy that believes it is incorrect can dispute it through the process in Section 9.6.

12. Accessibility & Reach

- WhatsApp as the primary channel: no app install, works on the phone people already have and already know how to use.
- Regional-language replies (Hindi, Kannada at launch; architecture supports adding more) for both the verdict text and the Safe Action guidance.
- Voice-note replies: the Safe Action text is also sent as a text-to-speech audio message in the chosen language, for users with low literacy or visual impairment — the group most exposed to unreadable fine print in the first place.
- High-contrast, large-tap-target UI on the web dashboard, designed for older users and low digital literacy, not just for a demo screenshot.
- Offline/low-bandwidth behaviour as described in Section 8 — a cached verdict beats no verdict.
- Future extension (not built for the hackathon, named as a roadmap item): a plain-SMS fallback for feature phones with no WhatsApp/data access at all, since Twilio's SMS API would work on the same infrastructure already in use for the WhatsApp bot.

13. Batch Timeline & Regulator Investigation View (P2 — stretch)

13.1 Batch Timeline

A chronological view for a given batch_number: Manufactured (mfg_date) → NSQ Alert, if any (date + reason) → aggregated scan events (count, first/last seen, geographic spread at district granularity — synthetic data in this build) → reports filed (dates, statuses, never showing raw counts publicly per Conflict Resolution rule 5) → disputes filed and resolved (Section 9.6). The data model already supports this; it is primarily a UI and query feature, which is why it is tiered P2 rather than core.

13.2 Regulator Investigation View

A separate, clearly-labeled prototype screen for pharmacists/regulators: all batches with open reports, a cross-manufacturer pattern view (a batch number repeating across unrelated packages, shown aggregated at district level — never raw scan-level geodata, to prevent the screen being used to infer a competitor's distribution footprint), the open dispute queue, and the ability to move a report through reports.status (open → under investigation → confirmed / dismissed). Access requires the prototype-grade token login described in Section 11.1 — the screen is explicitly labeled "prototype only; production would require real role-based access control" rather than pretending otherwise.

14. System Architecture

- **Client layer:** a web dashboard (scan / upload / manual entry, Evidence Matrix UI, Regulator view) and a WhatsApp bot (Twilio Sandbox for the hackathon; Twilio WhatsApp Business API as the production path).
- **API gateway / rate-limit layer:** sits in front of every endpoint; enforces the limits in Section 11.2 and rejects malformed uploads before they reach processing.
- **Auth layer:** issues and validates the report-identity, dispute-domain, and regulator tokens described in Section 11.1.
- **API layer:** FastAPI backend exposing /verify, /lookup-batch, /report, /batch-timeline, /disputes, /regulator, /user/data.
- **Processing layer:** QR/DataMatrix decoder, OCR extractor, the normalise()/fuzzy-match manufacturer resolver, and the Evidence Engine (decision logic + Conflict Resolution).
- **Data layer:** PostgreSQL (Supabase) — products, batches, nsq_alerts, scan_events (synthetic preview), scan_evidence, reports, disputes, consent_log, audit_log.
- **Caching layer:** a locally-cached last-known NSQ/spurious snapshot, refreshed on every successful ingestion run, serving the offline-mode behaviour in Section 8.
- **Ingestion layer:** a scheduled scraper/parser for CDSCO's NSQ and spurious-drug PDFs, polling once daily (not real-time, to be a polite, low-load client of a government site) with a descriptive User-Agent, backed by a hand-curated seed dataset as the fallback path — with a health-check job that alerts if zero new rows are ingested for more than 48 hours, so a silently broken scraper degrades to the seed set instead of going stale unnoticed.
- **Deployment:** Render (API) or Vercel (web client), connected to GitHub, with environment secrets for the WhatsApp integration and database connection — never committed to the repository.

Error handling & graceful degradation

- OCR confidence below threshold → the field is shown to the user for manual confirmation, never silently guessed.
- QR/DataMatrix fails to decode → the user is routed straight to manual entry with one tap, not left on a spinner.
- CDSCO ingestion fails or the source site changes structure → the API serves the last successfully cached snapshot with a visible "data as of [date]" notice, rather than returning an error or, worse, a false GREEN.
- WhatsApp API call fails → the API layer logs the failure and the client falls back to a plain-text response; the web dashboard remains fully functional independent of WhatsApp's availability.

15. Data Model

PostgreSQL schema. Types are given as a direct guide for a migration file — an AI coding assistant should be able to generate working SQL/ORM models straight from this table.

products

Field	Type	Notes
id	uuid, pk	
product_code	text	GTIN, unique
brand_name	text	
generic_name	text	
manufacturer	text	canonical name
manufacturer_alias_group_id	uuid, fk	links known spelling/format variants for fuzzy matching (Section 9.1)
licence_number	text	
created_at	timestampz	

manufacturer_aliases

Field	Type	Notes
id	uuid, pk	
alias_group_id	uuid	groups variant spellings of one real manufacturer
alias_text	text	e.g. "Cipla Ltd.", "CIPLA LIMITED", a transliterated form
is_canonical	boolean	

batches

Field	Type	Notes
id	uuid, pk	
product_code	text, fk → products	
batch_number	text	indexed
manufacturing_date	date	
expiry_date	date	drives the Section 9.4 expiry signal
manufacturer	text	
dispute_status	enum: none open resolved_upheld resolved_overturned	default 'none'

nsq_alerts

Field	Type	Notes
id	uuid, pk	
product_name	text	

Field	Type	Notes
batch_number	text	indexed
manufacturer	text	
alert_date	date	
reason	text	failure reason, verbatim from CDSCO
source_document	text	URL or filename of the ingested PDF
extraction_confidence	enum: high medium low	maps to Evidence Provenance confidence
lab_type	enum: central state	
status	enum: active corrected superseded	Conflict Resolution rule 4

scan_events (preview / synthetic data only)

Field	Type	Notes
id	uuid, pk	
hashed_serial_id	text	hashed, never the raw code — powers the replay/clone preview
product_code	text	
batch_number	text	
approx_location	text	pincode/district granularity only (Section 11.3)
scanned_at	timestampz	
source	enum: web whatsapp	
risk_level	enum: red amber expired green	
risk_reasons	text[]	

scan_evidence — powers the Evidence Matrix & Provenance

Field	Type	Notes
id	uuid, pk	
scan_id	uuid, fk	
signal_type	enum: code_validity label_consistency nsq_status expiry_status replay_pattern visual_heuristic community_reports	
status	text	Pass / FAIL / Inconclusive / EXPIRED
evidence_text	text	

Field	Type	Notes
source_type	enum: government_record decoded_code ocr_reading user_photo community_report	
source_reference	text	
confidence	enum: high medium low structured	
created_at	timestampz	

reports

Field	Type	Notes
id	uuid, pk	
scan_id	uuid, fk	
image_reference	text	photo evidence, subject to the retention policy in 11.4
report_type	text	
description	text	
reporter_identity_hash	text	hashed WhatsApp number or CAPTCHA token — never stored raw (DPDP data minimisation)
created_at	timestampz	
status	enum: open under_investigation confirmed dismissed	
visibility	enum: regulator_only	reports are never shown publicly — Conflict Resolution rule 5

disputes — NEW (Section 9.6)

Field	Type	Notes
id	uuid, pk	
batch_number	text, fk → batches	
submitted_by	text	manufacturer/pharmacy name
submitter_domain	text	email domain used for the hackathon-grade identity check (Section 11.1)
evidence_reference	text	uploaded proof document
status	enum: open under_review resolved_upheld resolved_overturned	
submitted_at	timestampz	

Field	Type	Notes
resolved_at	timestampz, nullable	
resolution_notes	text, nullable	

consent_log — NEW (DPDP compliance, Section 11.3)

Field	Type	Notes
id	uuid, pk	
user_identifier_hash	text	hashed phone number or session id
consent_type	enum: data_processing scan_history_opt_in location_opt_in	
granted_at	timestampz	
channel	enum: web whatsapp	

audit_log — NEW (security, Section 11.1)

Field	Type	Notes
id	uuid, pk	
actor	text	regulator token subject, or 'system'
action	text	e.g. 'report.status_changed'
target	text	record id affected
timestamp	timestampz	

16. API Specification

All endpoints are prefixed `/api`. Responses are JSON. Authenticated endpoints expect an `Authorization: Bearer <token>` header. This section is written to be implemented directly.

POST `/api/verify`

Public, rate-limited (Section 11.2). Accepts one of three input types.

```
// Request
{
  "input_type": "manual", // "manual" | "qr" | "photo"
  "batch_number": "AX2291",
  "product_code": "8901030812345", // optional if batch_number is globally unique
  "qr_payload": null, // present when input_type = "qr"
  "photo_base64": null, // present when input_type = "photo"; 10MB max,
  // JPEG/PNG/WebP
  "user_language": "hi" // ISO 639-1, default "en"
}

// Response – 200 OK
{
  "verdict": "RED", // "RED" | "AMBER" | "EXPIRED" | "GREEN"
  "verdict_reason": "nsq_match",
  "also_expired": false,
  "dispute_open": false,
  "evidence_matrix": [
    {
      "signal_type": "nsq_status",
      "status": "FAIL",
      "evidence_text": "Batch AX2291 listed Not of Standard Quality (dissolution
failure)",
      "source_type": "government_record",
      "source_reference": "CDSCO NSQ list, March 2026, Central Labs",
      "confidence": "structured"
    }
  ], // ...one object per row shown in Table 9.1
  "safe_action": {
    "text": "Do not consume. Show this result to a pharmacist. Keep the packaging.",
    "voice_note_url": "https://.../safe-action-hi-red.mp3",
    "report_enabled": true
  },
  "served_from_cache": false, // true when Section 8 offline mode is active
  "cache_last_updated": null
}
```

GET `/api/lookup-batch/{batch_number}`

Public, rate-limited. Returns the same shape as `/verify`'s response, for a direct-link/search use case without re-submitting a photo.

POST `/api/report`

Requires a reporting identity (Section 11.1), capped at 5/day/identity.

```
// Request
{
  "scan_id": "uuid",
  "report_type": "suspected_counterfeit",
  "description": "Packaging seal looked tampered.",
  "image_base64": null,
  "reporter_identity_token": "<whatsapp-number-token-or-captcha-token>"
}

// Response – 201 Created { "report_id": "uuid", "status": "open" }
```

GET /api/batch-timeline/{batch_number}

Public read. Returns manufacturing date, NSQ alert history, aggregated (district-level) scan-event summary, and dispute history — never raw report counts (Conflict Resolution rule 5).

POST /api/disputes

Requires the manufacturer-domain identity check (Section 11.1).

```
// Request
{
  "batch_number": "AX2291",
  "submitted_by": "Example Pharma Ltd.",
  "evidence_reference": "<uploaded document id>",
  "notes": "Attached corrected Certificate of Analysis."
}
// Response — 201 Created { "dispute_id": "uuid", "status": "open" }
```

GET /api/disputes/{batch_number}

Public read — lets the verdict screen show the "under review" banner (Section 9.4) without exposing internal regulator data.

POST /api/regulator/reports/{id}/status

Requires a regulator token (Section 11.1). Moves a report through open → under_investigation → confirmed / dismissed. Every call is written to audit_log.

DELETE /api/user/data

Requires the same identity check as /report — the person is deleting their own data (Section 11.3).

```
// Request { "identity_token": "<whatsapp-number-token>" }
// Response — 200 OK { "deleted": true, "scans_removed": 4, "photos_removed": 2 }
```

17. Screens & UX Specification

Written to be buildable without guessing. Each screen lists its purpose, layout, key states, and copy.

17.1 Home / Capture (web)

- Top bar: language selector (English / Hindi / Kannada), Hamsa logo.
- Three large, equally-weighted buttons: "Scan QR/DataMatrix", "Upload a Photo", "Enter Batch Number" — manual entry is never visually demoted to a small link.
- Below the buttons: a one-line disclaimer ("Not a substitute for professional medical advice") and a persistent DPDP consent notice on first visit.
- If the user has opted into scan history: a list of their last 5 scans with verdict-colored chips.

17.2 Verdict screen

- Full-width colored banner at the top matching the verdict (RED / AMBER / GREY for expired / GREEN), with the verdict name in large text.
- Safe Action text directly beneath the banner, in large, high-contrast type — this is the single most important line on the screen and must be visible without scrolling.
- If `dispute_open` is true: a distinct amber-outlined "Under review" banner appears between the verdict and the Safe Action text, never replacing either.
- Evidence Matrix below, as a vertical list of cards — one per signal — each showing an icon, the signal name, its status, and one line of evidence text, with a "Why?" affordance that expands into the full Provenance panel (source type, reference, confidence, last updated, dispute status).
- "Report this batch" and "Re-scan" buttons pinned at the bottom.
- If `served_from_cache` is true: a slim banner stating "Showing data as of [cache_last_updated] — reconnect to re-check."

17.3 Manual entry

- Single prominent batch-number field with an inline format hint (e.g. "e.g. AX2291").
- Optional product-name field with autocomplete against the products table.
- One submit button; on submit, routes straight to the Verdict screen.

17.4 WhatsApp flow

- First message ever: a one-time consent notice (Section 11.3) plus a language prompt (reply 1 for English, 2 for Hindi, 3 for Kannada).
- Main menu: reply with a batch number directly, or send a photo, or send the QR/DataMatrix photo — the bot auto-detects which.
- Verdict delivered as: a short structured text message (verdict, one-line reason, Safe Action text) followed by a voice-note audio message of the same Safe Action text.
- Every message footer: "Reply DELETE to erase your data."

17.5 Regulator Investigation View (prototype, auth-gated)

- Simple token-based login screen, clearly labeled "Prototype access — not production authentication."
- Table of open reports: batch number, report count, oldest open report age, status dropdown (open / under investigation / confirmed / dismissed).
- Cross-manufacturer pattern panel: batch numbers appearing across more than one distinct product listing, aggregated to district level.
- Dispute queue tab: open disputes with an approve/overturn action, each write recorded to `audit_log`.

17.6 Batch Timeline (P2)

A horizontal timeline component: Manufactured → NSQ Alert (if any) → aggregated scan activity → disputes filed/resolved. Read-only, public.

18. Technology Stack

Layer	Choice
Web frontend	React + Vite, Tailwind CSS, deployed on Vercel
Backend API	FastAPI (Python 3.11), deployed on Render
Database	PostgreSQL via Supabase
OCR	Tesseract as the primary, free, offline-capable engine; an optional hosted OCR API (e.g. Google Cloud Vision) only as a fallback for low-confidence results, explicitly costed in Section 23
QR / DataMatrix decode	pyzbar / zxing-cpp
Fuzzy matching	rapidfuzz (Python) against the manufacturer_aliases table
Messaging	Twilio WhatsApp Sandbox for the hackathon demo; Twilio WhatsApp Business API named as the production path
Rate limiting	slowapi (FastAPI) backed by a Redis instance, or an in-memory token bucket if Redis isn't provisioned in time
Auth	Signed short-lived JWTs for the regulator role; HMAC-signed tokens for report/dispute identity checks
Scraping / ingestion	BeautifulSoup + a scheduled job (APScheduler or a Render cron job), daily interval
Monitoring	A /health endpoint plus structured JSON logs; Sentry free tier for error tracking (optional)

19. Build Plan (34 Hours)

Deliberately scoped two hours short of a typical 36-hour window, as working buffer for setup, sleep, and rehearsal. Every item is tiered; P0/P1 items are the differentiator and are never cut.

Hours	Milestone	Tier	Fallback if behind
0-2	Repo + deploy skeleton, wireframes incl. Evidence Matrix and Verdict screen layout, DPDP consent-notice copy drafted	—	—
2-10	NSQ/spurious data pipeline: scrape and structure CDSCO PDFs, daily-poll scheduler, health-check alerting	P0	By hour 8, switch to a hand-curated seed set of ~50-100 real flagged batches
10-20	QR/DataMatrix decode + OCR + normalise()/fuzzy-match label-code check + expiry check + Evidence Engine logic + Conflict Resolution + scan_evidence logging	P0	By hour 16, if OCR accuracy is too low, promote manual batch entry to the visually primary input (it is already the logical default)
20-24	Security layer: rate limiting, EXIF stripping, report/dispute identity checks, regulator token auth, audit_log writes	P0	Cut the manufacturer-domain check to a manual allow-list if time-boxed

Hours	Milestone	Tier	Fallback if behind
24–28	Web dashboard: scan / upload / manual entry, Evidence Matrix, Provenance drill-down, Safe Action layer, dispute banner, scan history, offline-cache banner	P0/P1	Cut visual polish before ever cutting the Evidence Matrix
28–31	WhatsApp bot (Twilio Sandbox), Hindi/Kannada Safe Action text + voice-note replies, DELETE MY DATA command	P1	If setup stalls, demo a simulated WhatsApp-style chat inside the web app — and pre-send the sandbox join code to every judge's phone number before the demo slot, not during it
31–33	Stretch, in this order: (1) clone/replay preview, (2) manufacturer dispute submission flow, (3) Batch Timeline, (4) Regulator Investigation View — build as far down the list as time allows	P2	Regulator View is cut first, then Batch Timeline, then the dispute flow's UI (keep the API), then the clone/replay preview — in that order
33–34	Deploy, rehearse pitch + hardened Q&A, README + submission text, final QA against Section 20's checklist	—	—

20. Testing & QA Checklist

- QR/DataMatrix decode succeeds on the full prepared demo pack set (target $\geq 90\%$, Section 3.3).
- OCR extraction spot-checked on 20 sample packs; low-confidence fields correctly route to manual confirmation instead of being silently accepted.
- NSQ true-positive spot check: known NSQ batch numbers correctly return RED with the right source citation.
- NSQ true-negative spot check: a clean, in-date batch correctly returns GREEN.
- Expiry check: an expired-but-otherwise-clean batch returns the distinct Expired verdict, not RED.
- Conflict escalation unit test: two signals from the same evidence category do not escalate to RED; two from different categories do.
- Fuzzy-match unit test: known manufacturer alias pairs (e.g. "Ltd." vs "Limited") do not trigger a false label-consistency mismatch.
- Rate-limit trip test: 31st /verify call in a minute from one IP is correctly throttled.
- Auth-bypass attempt on /regulator/* without a token is correctly rejected.
- DPDP "DELETE MY DATA" command removes scan history and unattached photos, verified by a follow-up query.
- Offline-mode test: disabling network mid-session still returns a cached verdict with the correct "data as of" banner.
- WhatsApp sandbox join codes sent to every judge's phone number the day before the demo, with a rehearsed re-send step if a code expires.

21. Deployment Guide

Environment variables

- DATABASE_URL — Supabase Postgres connection string
- TWILIO_ACCOUNT_SID, TWILIO_AUTH_TOKEN, TWILIO_WHATSAPP_NUMBER
- JWT_SECRET — for regulator tokens

- `REPORT_IDENTITY_SECRET` — HMAC key for report/dispute identity tokens
- `OCR_FALLBACK_API_KEY` — only if the hosted OCR fallback is enabled
- `CDSKO_SCRAPER_INTERVAL_HOURS` — default 24

Setup steps

1. Provision a Supabase project; run the schema in Section 15 as a migration.
2. Deploy the FastAPI backend to Render, wiring the environment variables above.
3. Deploy the React web client to Vercel, pointing at the Render API URL.
4. Create a Twilio account, join the WhatsApp Sandbox, and configure the inbound webhook to the Render API's `/whatsapp` endpoint.
5. Run the ingestion job once manually to populate `nsq_alerts` from the seed dataset, then verify the scheduled daily job is registered.
6. Seed a small demo dataset: a known-RED batch, a known-expired batch, and a known-GREEN batch, matching the packs prepared for the live demo.
7. Send the WhatsApp Sandbox join code to every judge's phone number ahead of the demo slot.

22. Demo Script for Judges

1. **Real NSQ match** — enter/scan a genuinely NSQ-listed batch → show the Evidence Matrix with the RED row and the Safe Action text.
2. **Expired pack** — scan a clean but expired sample pack → show the distinct Expired verdict, and explicitly point out it is not the same UI as counterfeit RED — this is the technical-depth beat most teams miss.
3. **Label-vs-code mismatch** — scan a sample pack with an altered printed manufacturer → show the AMBER row, its Provenance drill-down, and the Safe Action text.
4. **Conflict escalation** — trigger two independent AMBER signals together → show them escalate to RED, and point out the Evidence Matrix visibly distinguishes this from an official NSQ match.
5. **Dispute flow** — submit a dispute against a demo batch → show the "under review" banner appear on the verdict screen without the underlying verdict silently vanishing.
6. **WhatsApp flow** — send a batch number to the bot → receive the Safe Action text as both a message and a voice note in Hindi or Kannada, no app install.
7. **Optional (P2)** — preview pattern check or Batch Timeline/Regulator View, shown only if built and always introduced verbally as "a preview" or "prototype-only" before the screen appears.

23. Business Model & Sustainability

23.1 Running cost, honestly estimated

Cost item	Approximate scale
Twilio WhatsApp Business API (post-sandbox)	Per-conversation pricing; free tier covers a hackathon demo, real usage needs a funded plan
Hosted OCR fallback (only for low-confidence Tesseract results)	A few dollars per 1,000 calls on typical cloud OCR pricing — kept small by using it only as a fallback, not the primary path
Render / Vercel / Supabase	Free tiers cover the hackathon and early pilot; paid tiers needed past roughly a few thousand monthly active users
Voice-note text-to-speech	Per-character/per-minute pricing on standard TTS APIs; small at pilot scale

23.2 Path to sustainability

The most credible long-term path is not consumer monetisation — a free safety tool that starts charging users undermines the exact trust it is built on. The realistic paths, roughly in order of likelihood, are: a state or central government licensing arrangement (the Karnataka and Kerala initiatives in Section 5 show real government appetite for exactly this kind of tool); a B2B API for pharmacy chains and distributors who want the Evidence Engine embedded in their own dispensing workflow; and CSR or public-health grant funding, given the direct alignment with patient-safety outcomes. This is stated as an honest, unresolved question, not a solved one — a judge who asks "who pays for this in year two" deserves a real answer, and pretending otherwise is a bigger risk than admitting it.

24. Honest Limitations — What We Are Not Claiming

- Not a genuineness certificate: GREEN means no warning was found in our data, not confirmed authenticity.
- Not forensic or lab-grade: no claim of detecting physical counterfeiting artifacts from a phone photo.

- The clone/replay weakness is real and only partly mitigated: a genuine, valid QR/DataMatrix code can be photographed and reprinted on a fake box, and the label-vs-code check will not catch it, because both sides agree. The preview clone/replay detector, running on synthetic data in this build, is a first step toward a real fix, not a solved problem — a production fix needs a live, manufacturer-integrated serialised scan-log, which requires direct manufacturer cooperation this hackathon build does not have.
- Batch Timeline and Regulator Investigation View are stretch goals, not guaranteed to ship; the Regulator View has prototype-grade authentication in this build, explicitly labeled as such.
- We are not the only ones working on this problem, and we say so: Kerala's MedWatch, Karnataka's live NSQ-freeze portal, MediCheck, and HealthSewa are disclosed in Section 5 rather than omitted.
- Consumer adoption of a bare verification tool is a real, evidenced risk — PharmaSecure's psVerify, a decade-old incumbent deployed on 1.2 billion packages, reportedly sees very little active mobile usage. Hamsa's WhatsApp-first, no-install design is a direct response to that risk, not a guarantee against it.
- "Hamsa" is a quick naming check, not trademark clearance — worth one more look before any submission beyond the hackathon.
- DPDP compliance as designed here (Section 11.3) covers the core principles a hackathon build can reasonably implement; a real production deployment would need a full compliance review against the DPDP Rules, 2025's phased obligations, not just this document.

25. Elevator Pitch (~90 seconds)

"Indian philosophy tells the story of the hamsa — a swan said to be able to drink the milk out of a mixture of milk and water, leaving the water behind. It's a classical symbol for discernment: telling the real from the false when both look the same. That's the exact problem with medicine in India today — a genuine pack and a counterfeit one can look identical, and a valid-looking QR code doesn't fix that, because codes can be copied too — we say that out loud instead of hiding it. Hamsa doesn't ask you to trust a single AI-generated score. Its Evidence Engine checks your medicine's batch against CDSCO's own monthly quality alerts — 189 batches flagged in a single month last year, including confirmed spurious drugs — alerts the government already publishes, but only as PDFs nobody searches. It checks whether the printed label matches what's encoded in the code, whether the medicine has even expired, and shows you an Evidence Matrix of exactly what was checked and where each fact came from — not a mystery number. If we get something wrong, a manufacturer can dispute it, and that dispute stays visible, not hidden. It reaches people over WhatsApp, in their own language, voice notes included, no app required — and it's built from day one to respect India's new data protection law, not bolt it on later. We're not claiming to replace a lab, and we're not the only team working on this problem — we checked, twice. We're claiming to turn evidence India already collects into an answer a patient can actually get, in seconds, for free, and to show our work while we do it."

26. Sources Referenced

- WHO — "1 in 10 medical products in developing countries is substandard or falsified," global surveillance report, 28 November 2017.
- Government of India, Drugs Rules 1945, Rule 96 / Schedule H2 — inserted by G.S.R. 823(E), 17 November 2022; in force from 1 August 2023 for the top 300 drug brands.
- Government of India, G.S.R. 506(E), Drugs (Seventh Amendment) Rules, 2026 — notified 22 June 2026; expands Schedule H2 to all vaccines, antimicrobials, NDPS drugs, and anti-cancer drugs, with staggered rollout (vaccines/anti-cancer/NDPS from 1 July 2027; antimicrobials from 1 July 2028).
- Lok Sabha unstarred question reply, 1 August 2025 — confirms the government has not disclosed how many manufacturers have adopted the QR-code mandate.
- Digital Personal Data Protection Act, 2023, and the Digital Personal Data Protection Rules, 2025 — Rules notified by MeitY on 13 November 2025.
- CDSCO public NSQ portal — cdscoonline.gov.in (monthly Not-of-Standard-Quality and spurious-drug lists, Central and State Laboratories).

- CDSCO monthly NSQ alert, June 2025 — 189 batches flagged (130 state, 55 central, 4 confirmed spurious) across 15 laboratories; Karnataka state labs accounted for 53.
- Karnataka Food Safety and Drug Administration — real-time NSQ-freeze and NDPS-tracking portal, launched May 2026 (Medical Dialogues).
- Kerala State Drugs Control Department — "MedWatch" app reported in development (mobileidworld.com); no confirmed public launch as of this writing.
- PharmaSecure psVerify — deployment and usage figures (company materials; AppGoblin third-party usage estimate).
- MediCheck — student hackathon prototype (Notion-hosted project page); HealthSewa — hackathon project listing (hackquest.io).

27. Pre-Submission Checklist

- All environment secrets are in .env, not committed to the repository.
- Demo data is seeded (Section 21) and matches the physical sample packs prepared for the live demo.
- WhatsApp Sandbox join codes sent to every judge ahead of the demo slot, with a rehearsed re-send step.
- The DPDP consent notice is visible on first use, on both web and WhatsApp.
- The dispute-submission contact path is live and tested end-to-end.
- Every disclaimer in Section 11.5 is visible in the actual UI, not just in this document.
- README written, covering setup steps from Section 21.
- Final read-through: every Evidence Matrix row shown in the demo has a real, non-empty source — nothing shown is a placeholder.